Congruences modulo 3 for the series with $A(A(A(x))) = (1+x)\,A(A(x))$

A proof of the conjecture in OEIS A396102, and of a family of similar statements

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Abstract

The OEIS entry A396102, contributed by Paul D. Hanna, concerns the power series $A(x) = x+x^2-2x^3+10x^4-71x^5+616x^6-\cdots$ determined by the equation $A(A(A(x))) = (1+x)\cdot A(A(x))$, which relates the second and third iterates of A (the series obtained by substituting A into itself). Hanna conjectured that every coefficient of positive degree leaves remainder 1 on division by 3. We prove this, as one case of a family of results: for every $r \geq 1$, the solution of $F^{\circ r} = (1+x)\cdot F^{\circ(r-1)}$ agrees modulo r with $x/(1-x) = x+x^2+x^3+\cdots$; the case $r = 3$ is Hanna's conjecture. The key step is to pass to the compositional inverse of the solution, which satisfies a simpler equation, and to observe that $x/(1+x)$ solves that equation modulo r. As by-products, every term of OEIS A140094 is 1 modulo 3, and every term of OEIS A067145 is odd. The main results have been checked by computer and proved formally in the Lean proof assistant.

How this paper was produced. This paper reports the result of an AI research effort directed by Dev D. Goyal. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning and working autonomously from prompts by Dev D. Goyal, found the proof, wrote its Lean 4 formalization, and wrote a first draft of the paper. Dev D. Goyal chose the problems and asked for proofs, formal verification and revisions, but did not supply proof ideas. Claude Opus 5.5, used with Claude Code, then checked the work independently: it rebuilt the formal proofs from source and confirmed that they rely only on Lean's three standard axioms, recomputed the sequence from its definition compared it with the published terms and tested every claim numerically, and read every step of the proofs. It also wrote the present version of this paper. Dev D. Goyal has reviewed the proofs and is responsible for them; this review is not an independent expert validation. The paper has not been peer reviewed. The section "About this paper" at the end gives the details.

1 Introduction

1.1 The sequence

The *On-Line Encyclopedia of Integer Sequences* (OEIS) [5] is a large public database of integer sequences. Many entries also record *conjectures*: patterns that the contributor observed in the data but did not prove.

This paper is about entry A396102 [1], contributed by Paul D. Hanna in June 2026. Its name is:

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G.f.  A(x) satisfies A(A(A(x))) = (1+x) * A(A(x)).
```

— OEIS A396102, Paul D. Hanna, June 2026

Here "G.f." stands for *generating function*: the series $A(x) = \sum_{n\geq 1} a(n)\, x^n$ whose coefficients are the terms of the sequence. The expressions $A(A(x))$ and $A(A(A(x)))$ are *iterates*: A substituted into itself once and twice (Section 2). The factor $(1+x)$ is an ordinary multiplication. So the equation says: substituting A into itself twice gives the same series as substituting it once and then multiplying by $1 + x$. Together with the normalization $A(x) = x + \cdots$, this determines A completely (Lemma 4.1). The first terms, from $n = 1$, are 1, 1, −2, 10, −71, 616, −6119, 67210, −799307, 10152043, −136403576, 1925530303, ...; the listed terms alternate in sign from $n = 2$ on.

The entry also records that the compositional inverse R of A (the series with $R(A(x)) = x$) satisfies the simpler equation $R(x) = x/(1+R(R(R(x))))$, and that $-R(-x)$ is the generating function of an older sequence, OEIS A140094 [2]. It compares A with the series of OEIS A067145 [3], which satisfies $F(F(x)) = (1 + x)F(x)$.

1.2 The question

The entry contains the following conjecture.

> `Conjecture: a(n) == 1 (mod 3) for n >= 1.`
>
> — OEIS A396102, Paul D. Hanna, June 2026

Here `a(n) == 1 (mod 3)` means that $a(n)$ leaves remainder 1 on division by 3. The table shows the remainders of the first terms modulo 3, and, for comparison, modulo 9. Modulo 3 every remainder is 1; modulo 9 the remainders are 1, 4 or 7 (the only possibilities, given the remainder modulo 3) without an evident pattern.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$a(n)$ mod 3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
$a(n)$ mod 9	1	1	7	1	1	4	1	7	1	7	1	4	7	7	7	4

n	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
$a(n)$ mod 3	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
$a(n)$ mod 9	7	7	1	7	4	7	4	7	1	4	4	4	1	4	4	7

1.3 What this paper proves

We study the whole family of equations

$$F^{\circ r}(x) = (1 + x) \cdot F^{\circ(r-1)}(x), \qquad F(x) = x + \cdots \qquad (r \geq 1), \tag{1}$$

where $F^{\circ r}$ is the r-th iterate of F. The case $r = 3$ is the equation of A396102, and the case $r = 2$ is the equation $F(F(x)) = (1 + x)F(x)$ of A067145.

Main result (Theorem 3.1). *For every $r \geq 1$, equation* (1) *has exactly one solution F_r with integer coefficients, and modulo r*

$$F_r(x) \equiv \frac{x}{1-x} = x + x^2 + x^3 + \cdots \pmod{r}.$$

For every $n \geq 1$, the coefficient of x^n in F_r is congruent to 1 modulo r. The coefficient of x^n in the k-th iterate of F_r is congruent to k^{n-1} modulo r.

For $r = 3$ this proves Hanna's conjecture (Corollary 3.2). We also deduce that every term of A140094 is 1 modulo 3 and every term of A067145 is odd (Section 6).

1.4 What was known

The equation for the inverse series R recorded in the entry (formula (5) in its list of formulas) is the starting point of our proof; we prove it from the definition (Lemma 4.1). In October 2026, Amol Bhave added to OEIS A196523 [4] a proof of a congruence for a different family: for every $\ell \geq 2$, the coefficients of the solution of $A(x) = x + x \cdot A^{\circ \ell}(x)$, where $A^{\circ \ell}$ is the ℓ-th iterate of A, are all 1 modulo $\ell - 1$. The proof compares the solution with $x/(1-x)$, as we do here. The equation treated here is different: it relates two consecutive iterates, and the comparison works for the inverse series rather than for A itself. We are not aware of an earlier proof of the conjecture of A396102.

1.5 Motivation

Equations that involve iterates of an unknown series, such as $A(A(A(x))) = (1+x)A(A(x))$, are hard to study: there is no general method for solving them, and the coefficients of the solutions are usually large and irregular. In this example the remainders are explained by a single series, $x/(1-x)$, whose iterates are $x/(1-kx)$. The proof works with a whole family of equations at once, and with the inverse series rather than with A; it also gives results for two other OEIS entries.

1.6 How to read this paper

Section 2 explains the few notions we need: power series, substitution, compositional inverses, remainders, and the series $x/(1-cx)$. Section 3 states the results precisely, and Section 4 proves them. Section 5 describes how the results were checked, by computer and by a formal proof in Lean. Section 6 discusses consequences for two related OEIS entries and some open questions.

2 Background

This section collects the definitions and facts used later. Readers who know formal power series and congruences can skip it and return when needed.

2.1 Formal power series

A *formal power series* is an expression

$$U(x) = u_0 + u_1 x + u_2 x^2 + u_3 x^3 + \cdots,$$

determined by its sequence of coefficients $u_0, u_1, u_2, \ldots$; see [7, 6]. No number is ever substituted for x, so questions of convergence do not arise. We write $[x^n]\,U = u_n$ for the coefficient of x^n in U; two series are equal when all their coefficients are equal.

Series are added coefficient by coefficient and multiplied like polynomials:

$$[x^n]\,(UV) = u_0 v_n + u_1 v_{n-1} + \cdots + u_n v_0.$$

Each coefficient of a product is a finite sum. For example, $(1-x)(1+x+x^2+x^3+\cdots) = 1$, so $1 + x + x^2 + \cdots$ is the *multiplicative inverse* of $1 - x$, written $\frac{1}{1-x}$. More generally, every series D with constant term 1 has a unique multiplicative inverse $\frac{1}{D}$, whose coefficients are

determined one at a time by $D \cdot \frac{1}{D} = 1$ using only addition, subtraction and multiplication. Whenever a quotient of power series appears in this paper, its denominator has constant term 1, so the quotient is again a power series.

A *commutative ring* is a set with an addition, a subtraction and a multiplication that satisfy the usual rules of arithmetic; examples are $\mathbb{Z}$, $\mathbb{Q}$ and the integers modulo m. For a commutative ring R we write $R[[x]]$ for the set of power series with coefficients in R; for instance, writing $\mathbb{F}_2$ for the integers modulo 2, $\mathbb{F}_2[[x]]$ consists of the series with coefficients modulo 2. We write $U(0)$ for the constant term u_0 of U, and $U = O(x^d)$ when the coefficients of $1, x, \ldots, x^{d-1}$ in U are all zero. Thus $U - V = O(x^d)$ means that U and V have the same coefficient of x^n for every $n < d$. A series with constant term 0 and coefficient of x equal to 1, that is, a series of the form $x + u_2x^2 + u_3x^3 + \cdots$, is called *normalized*.

2.2 Substituting one series into another

If $V(x)$ has constant term 0, we can substitute V for x in U:

$$U(V(x)) = u_0 + u_1V(x) + u_2V(x)^2 + u_3V(x)^3 + \cdots .$$

This is called *composition*, also written $U \circ V$. It makes sense because $V(x)^k$ has no terms below x^k, so only the finitely many terms with $k \le n$ affect the coefficient of x^n. For example, if $U = x+x^2$ and $V = x^2$, then $U(V(x)) = x^2+x^4$. Composition is associative: $U \circ (V \circ W) = (U \circ V) \circ W$.

Substituting a series into itself repeatedly gives its *iterates*: $U^{\circ 2}(x) = U(U(x))$, $U^{\circ 3}(x) = U(U(U(x)))$, and so on, with $U^{\circ 0}(x) = x$ and $U^{\circ 1} = U$. They must not be confused with *powers* $U^2 = U \cdot U$, $U^3 = U \cdot U \cdot U$. (The OEIS sometimes writes $A^k(x)$ for the k-th iterate. In this paper the small circle always means an iterate, and a plain exponent always means a power.)

2.3 Undoing a substitution

A normalized series can always be "undone" by another normalized series.

Lemma 2.1 (Compositional inverse)**.** *For every normalized series U there is exactly one normalized series V with $U(V(x)) = x$, and this V also satisfies $V(U(x)) = x$. If U has integer coefficients, so does V.*

We call V the *compositional inverse* of U. The OEIS calls it the *series reversion* of U. For example, the compositional inverse of $x - x^2$ is $C(x) = x + x^2 + 2x^3 + 5x^4 + 14x^5 + \cdots$, whose coefficients are the Catalan numbers. Indeed $C - C^2 = x$, which says exactly that substituting C into $x - x^2$ gives x.

Proof. Write $V = x + v_2x^2 + v_3x^3 + \cdots$ with unknown coefficients. In $U(V) = V + u_2V^2 + u_3V^3 + \cdots$, the coefficient of x^n (for $n \ge 2$) equals v_n plus terms that involve only $v_2, \ldots, v_{n-1}$. Setting it equal to 0 therefore determines v_n from the earlier coefficients, using only additions, subtractions and multiplications. So there is exactly one V with $U \circ V = x$, and it has integer coefficients if U does. Applying the same construction to V gives a normalized W with $V \circ W = x$. Then $U = U \circ (V \circ W) = (U \circ V) \circ W = W$, so $V \circ U = V \circ W = x$ as well. □

2.4 Remainders and congruences

For an integer $m \geq 2$ we write $a \equiv b \pmod{m}$ ("a is congruent to b modulo m") when $a - b$ is divisible by m, that is, when a and b leave the same remainder after division by m. The remainder of a on division by m is called its *residue* modulo m; it is one of $0, 1, \ldots, m-1$.

Addition and multiplication are compatible with congruences, so the residues $0, 1, \ldots, m-1$, added and multiplied modulo m, form a ring, written $\mathbb{Z}/m\mathbb{Z}$. When $m = p$ is a prime we also write $\mathbb{F}_p$. In $\mathbb{F}_2 = \{0, 1\}$ we have $1 + 1 = 0$, so addition and subtraction coincide.

For power series with integer coefficients, $U \equiv V \pmod{m}$ means that *every* coefficient of $U - V$ is divisible by m. Equivalently, the two series become equal when every coefficient is replaced by its residue; we call this *reducing* the series modulo m. The key fact is the following.

Reduction modulo m is compatible with all the operations above. To reduce a sum, a product, a substitution, or an inverse, we may reduce the ingredients first and then compute in $\mathbb{Z}/m\mathbb{Z}$. This holds because every coefficient of the result is obtained from finitely many coefficients of the ingredients by adding, subtracting and multiplying, and these operations are compatible with remainders.

Rings. Some statements below hold over every commutative ring, for example $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{Z}/m\mathbb{Z}$. No division is used in them; this matters because in $\mathbb{Z}/m\mathbb{Z}$ division is not always possible (for instance, 2 has no inverse in $\mathbb{Z}/4\mathbb{Z}$).

2.5 The method

The main proofs follow three steps.

1. *Uniqueness.* Show that the defining equation determines the coefficients one at a time, using only addition, subtraction and multiplication. Then the equation has exactly one solution of the required shape, in every commutative ring, including $\mathbb{Z}/r\mathbb{Z}$.
2. *Reduction.* Reducing the integer solution modulo r gives a solution of the reduced equation, because reduction is compatible with all operations.
3. *A candidate solution.* Find an explicit series that satisfies the reduced equation. By uniqueness, it is the reduction of the integer solution, and its coefficients can be read off.

Finding the candidate in step 3 is usually the main difficulty.

2.6 The series $x/(1-cx)$

For a number c in a commutative ring, put

$$M_c(x) = \frac{x}{1-cx} = x + cx^2 + c^2x^3 + c^3x^4 + \cdots = \sum_{n \geq 1} c^{n-1}x^n.$$

For $n \geq 1$, the coefficient of x^n in M_c is c^{n-1} (with $c^0 = 1$). For example $M_1 = x/(1-x) = x + x^2 + x^3 + \cdots$ has every coefficient of positive degree equal to 1, $M_0 = x$, and $M_{-1} = x/(1+x) = x - x^2 + x^3 - \cdots$.

Lemma 2.2. *For all c, d in a commutative ring, $M_c(M_d(x)) = M_{c+d}(x)$. Consequently the k-th iterate of M_c is M_{kc}, and M_{-c} is the compositional inverse of M_c.*

Proof. Substituting $y = M_d(x) = x/(1-dx)$ into $M_c(y) = y/(1-cy)$ and multiplying the numerator and the denominator by $1-dx$ gives

$$M_c(M_d(x)) = \frac{x}{(1-dx)-cx} = \frac{x}{1-(c+d)x} = M_{c+d}(x).$$

(All denominators have constant term 1, so these fractions are power series.) Applying this $k-1$ times gives $M_c^{\circ k} = M_{kc}$, and $M_c(M_{-c}(x)) = M_0(x) = x$. □

3 Results

Theorem 3.1. *For every integer $r \geq 1$, equation* (1) *has exactly one solution F_r among the normalized power series with integer coefficients, and*

$$F_r \equiv \frac{x}{1-x} \pmod{r}.$$

More generally, for all $k \geq 0$ and $n \geq 1$, the coefficient of x^n in the k-th iterate satisfies

$$[x^n]\, F_r^{\circ k} \equiv k^{\,n-1} \pmod{r}.$$

Corollary 3.2 (Hanna's conjecture)**.** *The series A of OEIS A396102 is F_3. Hence $a(n) \equiv 1 \pmod 3$ for every $n \geq 1$, and the coefficients of the k-th iterate of A satisfy $[x^n]\, A^{\circ k} \equiv k^{\,n-1} \pmod 3$.*

For instance, the second iterate $A(A(x)) = x+2x^2-2x^3+11x^4-80x^5+\cdots$, listed in the entry, has coefficients $\equiv 2^{n-1} \pmod 3$, that is, $1, 2, 1, 2, 1, \ldots$; and the third iterate $A(A(A(x))) = x + 3x^2 + 0x^3 + 9x^4 - 69x^5 + \cdots$ has all coefficients after the first divisible by 3.

4 Proofs

The idea in brief. Equation (1) is awkward to work with directly. Its compositional inverse R satisfies the simpler equation $R = x - R \cdot R^{\circ r}$ (Lemma 4.1), which determines R one coefficient at a time. Modulo r, the series $x/(1+x)$ solves this equation: its r-th iterate is $x/(1+rx)$, which is x modulo r. So, by uniqueness, $R \equiv x/(1+x)$, and therefore F_r, the inverse of R, is congruent to $x/(1-x)$.

Lemma 4.1. *Let $r \geq 1$, and work in any commutative ring.*

(a) There is exactly one power series R with constant term 0 that satisfies

$$R = x - R \cdot R^{\circ r}, \tag{2}$$

and it is normalized.

(b) A normalized series F satisfies (1) *if and only if its compositional inverse satisfies* (2)*.*

Consequently (1) *has exactly one normalized solution, namely the compositional inverse of R, and reducing the integer solution modulo r gives the solution over $\mathbb{Z}/r\mathbb{Z}$.*

Proof. (a) We show that the coefficient of x^n in $R \cdot R^{\circ r}$ depends only on the coefficients of R below degree n. Then (2) determines R one coefficient at a time, using only additions,

subtractions and multiplications, so it has exactly one solution. Since $R \cdot R^{\circ r}$ has no terms below x^2, the solution starts with x, that is, it is normalized.

Suppose G and H have constant term 0 and agree below degree N, that is, $G - H = O(x^N)$. Then $G(U) - H(U) = (G - H)(U) = O(x^N)$ for every series U with constant term 0, and $H(U) - H(V) = O(x^N)$ whenever $U - V = O(x^N)$, because each $U^k - V^k$ is divisible by $U - V$. Applying these two facts repeatedly shows that all iterates of G and H agree below degree N. Then

$$G \cdot G^{\circ r} - H \cdot H^{\circ r} = G\left(G^{\circ r} - H^{\circ r}\right) + (G - H)\, H^{\circ r} = O(x^{N+1}),$$

because in each product one factor is $O(x^N)$ and the other has no constant term. Taking $N = n$ gives the claim.

(b) Let F be normalized with inverse R, so that $R(F(x)) = x$ and $F(R(x)) = x$ (Lemma 2.1). Suppose R satisfies (2). Substitute $F^{\circ r}(x)$ for x in (2). Since $R \circ F^{\circ r} = F^{\circ(r-1)}$ and $R^{\circ r} \circ F^{\circ r} = x$, this gives

$$F^{\circ(r-1)} = F^{\circ r} - F^{\circ(r-1)} \cdot x,$$

which is (1). Conversely, suppose F satisfies (1), and substitute $R^{\circ r}(x)$ for x. Since $F^{\circ r} \circ R^{\circ r} = x$ and $F^{\circ(r-1)} \circ R^{\circ r} = R$, this gives $x = (1 + R^{\circ r}) \cdot R$, which is (2). (Substitution is compatible with sums and products, so we may substitute into both sides of an identity.)

Finally, by (a) and (b) the normalized solutions of (1) are exactly the inverses of the one solution of (2), so there is exactly one. Reduction modulo r is compatible with all the operations involved, so it maps the integer solution of (2) to the solution over $\mathbb{Z}/r\mathbb{Z}$, and it maps compositional inverses to compositional inverses. □

Proof of Theorem 3.1. Work over $\mathbb{Z}/r\mathbb{Z}$ and use the series $M_c = x/(1 - cx)$ of Lemma 2.2. We check that $M_{-1} = x/(1 + x)$ solves (2). By Lemma 2.2, $M_{-1}^{\circ r} = M_{-r}$, and $-r = 0$ in $\mathbb{Z}/r\mathbb{Z}$, so $M_{-1}^{\circ r} = M_0 = x$. Therefore

$$x - M_{-1} \cdot M_{-1}^{\circ r} = x - \frac{x}{1+x} \cdot x = \frac{x(1+x) - x^2}{1+x} = \frac{x}{1+x} = M_{-1}.$$

By the uniqueness in Lemma 4.1, the reduction of R modulo r is M_{-1}. Its compositional inverse is M_1 (Lemma 2.2), so $F_r \equiv M_1 = x/(1 - x) \pmod r$. Finally, reduction is compatible with substitution, so $F_r^{\circ k} \equiv M_1^{\circ k} = M_k = \sum_{n \geq 1} k^{\,n-1} x^n \pmod r$. □

Proof of Corollary 3.2. The series A is normalized and satisfies (1) with $r = 3$, so $A = F_3$ by the uniqueness in Lemma 4.1. Apply Theorem 3.1 with $r = 3$. □

5 Checking the results

A mathematical proof can contain mistakes, so the results were checked in two independent ways.

Numerical check. We solved the defining equation coefficient by coefficient, with exact integer arithmetic. This reproduces all 23 terms printed in the OEIS entry. The congruence $a(n) \equiv 1 \pmod 3$ holds for every $n < 70$, and so does $[x^n]\, A^{\circ k} \equiv k^{\,n-1} \pmod 3$ for $0 \leq k \leq 8$. For the family, we checked that every coefficient of F_r is 1 modulo r, for $2 \leq r \leq 7$ and $n < 40$. The computation was done with the PARI/GP computer algebra system, using code that was written separately from the formal proofs (file `verification/independent_checks.gp` in the accompanying repository [10]). A computation can only test finitely many cases, so it is a check on the definitions and on the indexing, not a proof.

Formal proof in Lean. Lean [8] is a *proof assistant*: a program in which mathematical statements and their proofs are written in a precise formal language, and every step is checked mechanically down to the basic rules of logic. The theorems of Section 3 are proved in Lean 4 (version 4.35.0) on top of Mathlib [9], Lean's library of formalized mathematics; the steps proved only on paper are identified below. The statements are collected in the file `lean/Entries/A396102.lean` of the repository, in the namespace `OEIS.A396102`. In Lean, `family_source_existence r` and `family_source_equation r` state the case $r+1$ of (1), so that $r = 0$ in Lean corresponds to $r = 1$ in this paper; the other declarations use the same r as the paper. The proofs are in the file `lean/Families/InverseIterate.lean`. The consequences for OEIS A140094 and A067145 in Section 6 are not part of the Lean formalization. The table lists the results covered by the Lean formalization. The Lean formalization does not cover the additional arguments and observations in Section 6.

Statement in this paper	Lean theorem(s)
A is normalized, satisfies $A(A(A(x))) = (1+x)A(A(x))$, and is the only such series	`normalized`, `source_equation`, `source_unique`, `source_existence_uniqueness`
Corollary 3.2: $a(n) \equiv 1 \pmod 3$, and the iterates	`coefficient_congruence`, `iterate_congruence`
Lemma 4.1: the family (1) and the equation (2) for the inverse	`family_source_existence`, `family_source_equation`, `auxiliary_source_equation`, `inverse_right`, `inverse_left`
Theorem 3.1 for every r	`family_coefficient_congruence`, `family_iterate_congruence`

The Lean project was rebuilt from source, and the axioms on which each of these theorems depends were listed with Lean's `#print axioms` command. Every one depends only on the three standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`); no step is left unproved. A formal proof certifies exactly the statement that Lean checks. Whether that statement says what the OEIS entry means is a separate question, so the Lean definitions were also compared with the definition in the OEIS entry (Section 1.1). The principal theorems are stated once more, using only Mathlib's definitions, in the file `lean/Challenges/FurtherResults.lean`, and Lean's Comparator tool confirms that the proved theorems have exactly these statements.

6 Discussion

Scope. Theorem 3.1 determines every coefficient of every member of the family modulo r, and every coefficient of every iterate. It says nothing about higher powers of r: modulo 9 the coefficients of A396102 show no evident pattern (see the table in the Introduction). It also says nothing about the signs. The computed terms $a(2), a(3), \ldots, a(59)$ of A396102 alternate in sign; we have not proved that this continues.

Two related entries. The theorem has consequences for two older OEIS entries, which we checked against their published terms.

- *OEIS A140094* [2], by Paul D. Hanna (2008), has a generating function S with $S(x) = x/(1 - S(S(S(x))))$. Let R be the inverse of A as above, and put $S(x) = -R(-x)$, as recorded in A396102. From $R(-y) = -S(y)$ one gets $R^{\circ 3}(-x) = -S^{\circ 3}(x)$, and

substituting $-x$ for x in $R = x - R \cdot R^{\circ 3}$ gives $S = x + S \cdot S^{\circ 3}$, which is the same as $S = x/(1 - S^{\circ 3})$. (By the argument of Lemma 4.1(a), this equation has only one solution with constant term 0, so S is indeed the generating function of A140094.) Since $R \equiv x/(1 + x) \pmod 3$, we get $S(x) \equiv x/(1 - x) \pmod 3$. So *every term of A140094 is* 1 *modulo* 3. The 23 terms printed in the entry agree with this. The same argument shows that, for every r, the solution of $S = x/(1-S^{\circ r})$ is congruent to $x/(1-x)$ modulo r.

- *OEIS A067145* [3], by Christian G. Bower (2002), is the sequence that "shifts left under reversion": its generating function F satisfies $R(x) = F(x)/x - 1$, where R is its inverse. Substituting $F(x)$ for x turns this into $F(F(x)) = (1 + x)F(x)$, as noted in that entry; this is (1) with $r = 2$. So $F = F_2$, and by Theorem 3.1 *every term of A067145 is odd.* The 23 terms printed in the entry agree with this.

Why $x/(1 - x)$? The series $M_1 = x/(1 - x)$ is special because its iterates have the explicit form $M_1^{\circ k} = x/(1 - kx)$. Modulo r, the r-th iterate is x again. So in any equation in which the number of iterations matters only modulo r, the series $x/(1 - x)$ or $x/(1 + x)$ is a natural candidate for a solution modulo r. The same observation underlies Bhave's proof for A196523 [4]. It would be interesting to know which other iterative equations in the OEIS can be treated this way.

About this paper

How the work was done. This paper is one of 28 papers on conjectures from the OEIS that came out of an AI research effort directed by Dev D. Goyal [10]. The roles were as follows.

- *Dev D. Goyal* started the project, chose which problems to pursue, asked for proofs, formal verification and revisions, and decided what to publish. Dev D. Goyal did not supply proof strategies or intermediate arguments.
- *ChatGPT* (OpenAI), using the GPT-6 model with Pro reasoning, worked autonomously from those prompts. It searched the OEIS for suitable conjectures, found the proofs, wrote the Lean 4 formalization, and wrote a first draft of this paper.
- *Claude Opus 5.5*, used with Claude Code, checked the work independently. It rebuilt the Lean project from source and confirmed that the theorems listed in Section 5 rely only on Lean's standard axioms; compared the Lean definitions with the OEIS definition; recomputed the sequence with separately written PARI/GP code, compared it with the terms published in the OEIS, and tested every claim of the paper on many terms; and read every step of the proofs. It then wrote the present version of the paper, expanding the explanations and, in places, writing out an argument in more detail or by a more elementary route. The theorems of Section 3 and the proof strategies are ChatGPT's. The Discussion section, including the short arguments in it, was written by Claude Opus 5.5; its observations were checked by computer and are not part of the Lean formalization.

Dev D. Goyal has reviewed the proofs and is responsible for this paper; this review is not an independent expert validation.

Review status. This paper has not been peer reviewed, and it has not been submitted to a journal or to the OEIS.

Priority. An OEIS entry that labels a statement as a conjecture shows only that its contributor did not know a proof; it does not show that no proof existed. During the research only limited searches for earlier work were made. No claim of priority is made.

Corrections. Please report any error in the issue tracker of the repository [10], where the Lean sources and the numerical checks are also available.

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